AI EXPLAINED SIMPLY

Token Cheat Sheet: How AI Reads Your Words

What a token is

- A token is a chunk of text: a whole word, part of a word, a space plus a word, or a punctuation mark.
- Before a model reads your message, a **tokenizer** splits it into tokens and turns each one into a number (an ID).
- Example (GPT-4 tokenizer, `cl100k_base`): `strawberry` → `str` + `aw` + `berry` → IDs 496, 675, 15717.
- That is why models have historically struggled with letter-level tasks like counting the R's in "strawberry": they see chunks, not letters.

Rules of thumb (English)

- 1 token ≈ 4 characters.
- 1 token ≈ ¾ of a word.
- 100 tokens ≈ 75 words.
- These are estimates. Other languages have different ratios.

Why tokens matter to you

1. **Memory (context window):** a model can only see a fixed number of tokens at once. GPT-4o: 128,000 tokens, about 96,000 English words. Text beyond the limit is not seen.
2. **Cost:** API usage is billed per token, input and output.
3. **Language:** the same idea can cost far more tokens in some languages. "I love pizza." = 4 tokens in the GPT-4 tokenizer; the same sentence in Hindi (मुझे पिज़्ज़ा पसंद है।) = 26 tokens. The newer GPT-4o tokenizer (`o200k_base`) brings the Hindi sentence down to 11.

Count tokens yourself (free)

- OpenAI's web Tokenizer tool: platform.openai.com/tokenizer
- In Python:

```
pip install tiktoken
python -c "import tiktoken; e=tiktoken.get_encoding('cl100k_base'); print(l
```

3 habits that save tokens

1. Paste only the part of a document the AI needs, not the whole file.
2. Start a fresh chat for a new task: long chats fill the context window.
3. Ask for a length ("in 5 bullet points"): output tokens count too.

Sources

- OpenAI Help Center, "What are tokens and how to count them": help.openai.com
- OpenAI model page, GPT-4o (128,000-token context window): developers.openai.com
- tiktoken, OpenAI's open-source tokenizer (token counts measured on 2026-11-11): github.com/openai/tiktoken
- TechCrunch, "Why AI can't spell 'strawberry'" (2024): techcrunch.com

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